

Agentic-Native 增长： Zilliz 如何用 AI Agent 支 撑超线性业务扩张

演讲人： 郭人通

Zilliz 联创，产品VP

AiCon

全球人工智能开发与应用大会

极客邦科技 2026 年会议规划

促进软件开发及相关领域知识与创新的传播



参会咨询

InfoQ 技术大会

QCon

上海站

全球软件开发大会

大模型正在重新定义软件

Large Language Model Is Redefining
The Software

10月22-24日 / 上海建工浦江皇冠假日酒店

· AI Agent

· Vibe Coding

· 智能可观测

· 推理基建

· 模型攻防

· AI x 创造力

InfoQ 技术大会

AiCon

北京站

全球人工智能开发与应用大会

探索 AI 应用边界

Explore the limits of AI applications

12月18-19日 / 北京

· 大模型架构创新

· 多模态 AI 产业融合

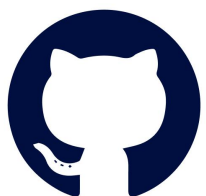
· 具身智能

· AI for Science

· 大模型安全

Milvus: 全球最受欢迎的开源向量数据库

广泛部署于全球近万家头部企业客户
企业级向量数据库解决方案，高性能、高可扩展、高度稳定
从原型构建到大规模生产部署，稳定支持百亿级向量规模



 300
+
contributors

 45K+
stars

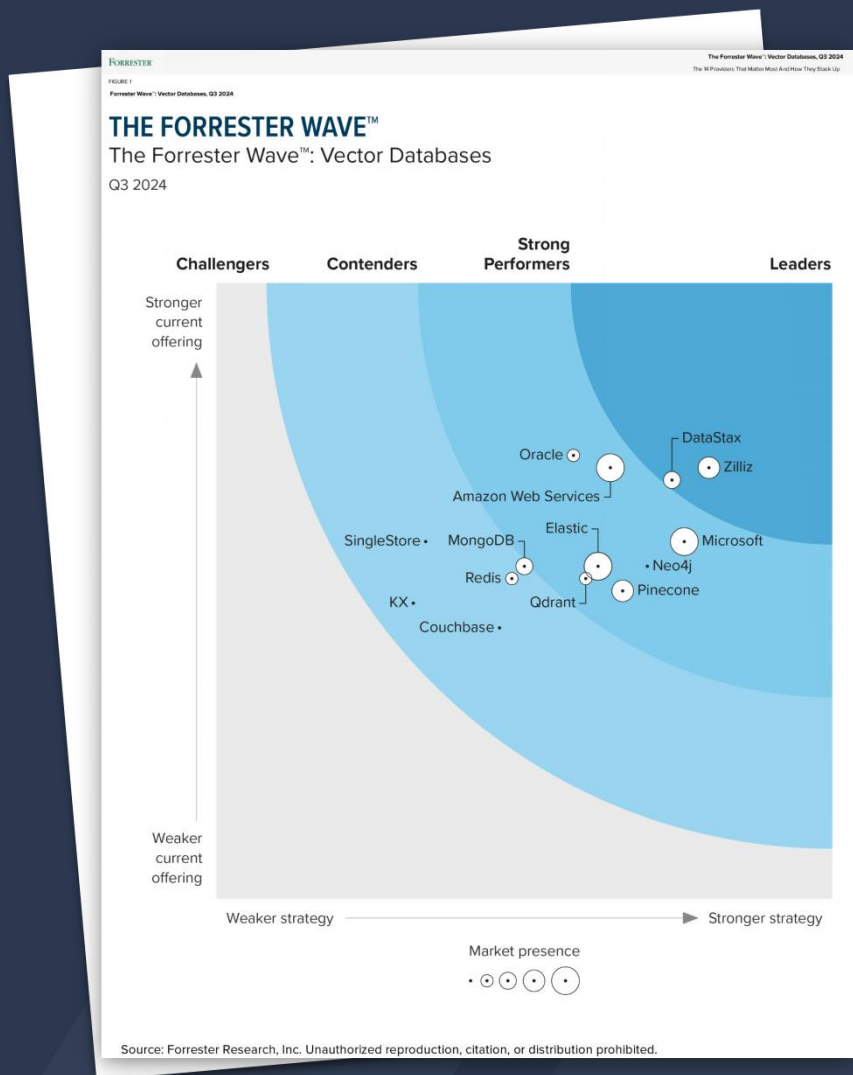
 150M+
active pods

 4.2K
+
forks

Zilliz Cloud: 向量数据库领导者象限全球第一

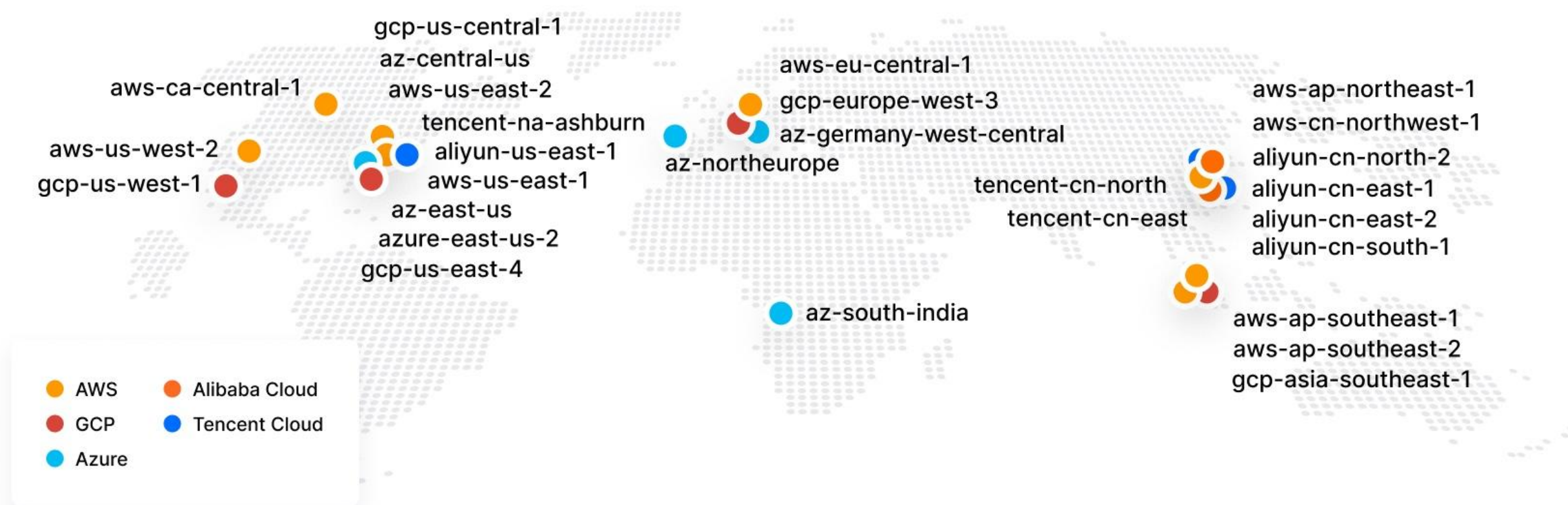
The Forrester Wave™ Vector Database Providers

Zilliz 在向量数据库领域深耕八年，陪伴全球客户深度迭代解决方案。



覆盖全球5朵公有云

Secure and compliant, globally distributed vector database infrastructure



深受企业用户信赖

 NVIDIA

Walmart 

 reddit

 Meta

 salesforce

ebay

Tencent

 ByteDance

Baidu  百度

 蚂蚁集团
ANT GROUP

 美团

 NI

 滴滴
A 站一下 又 站出

VOLVO

 MINIMAX

oppo

vivo

 理想

 NIO

CATL

 快手

小红书



iQIYI

 招商银行
CHINA MERCHANTS BANK

 中国建设银行
China Construction Bank

 华泰证券
HUATAI SECURITIES

 China
unicom 中国联通

 得物

 WPS

 贝壳
链家大平台

 SF TECHNOLOGY
顺丰科技

 喜马拉雅

 芒果tv

Transsion

 BOSCH

 exa

 Airtable

ROBLOX

 Filevine

 Read

OpenEvidence 

 doximity

 POSHMARK

notta

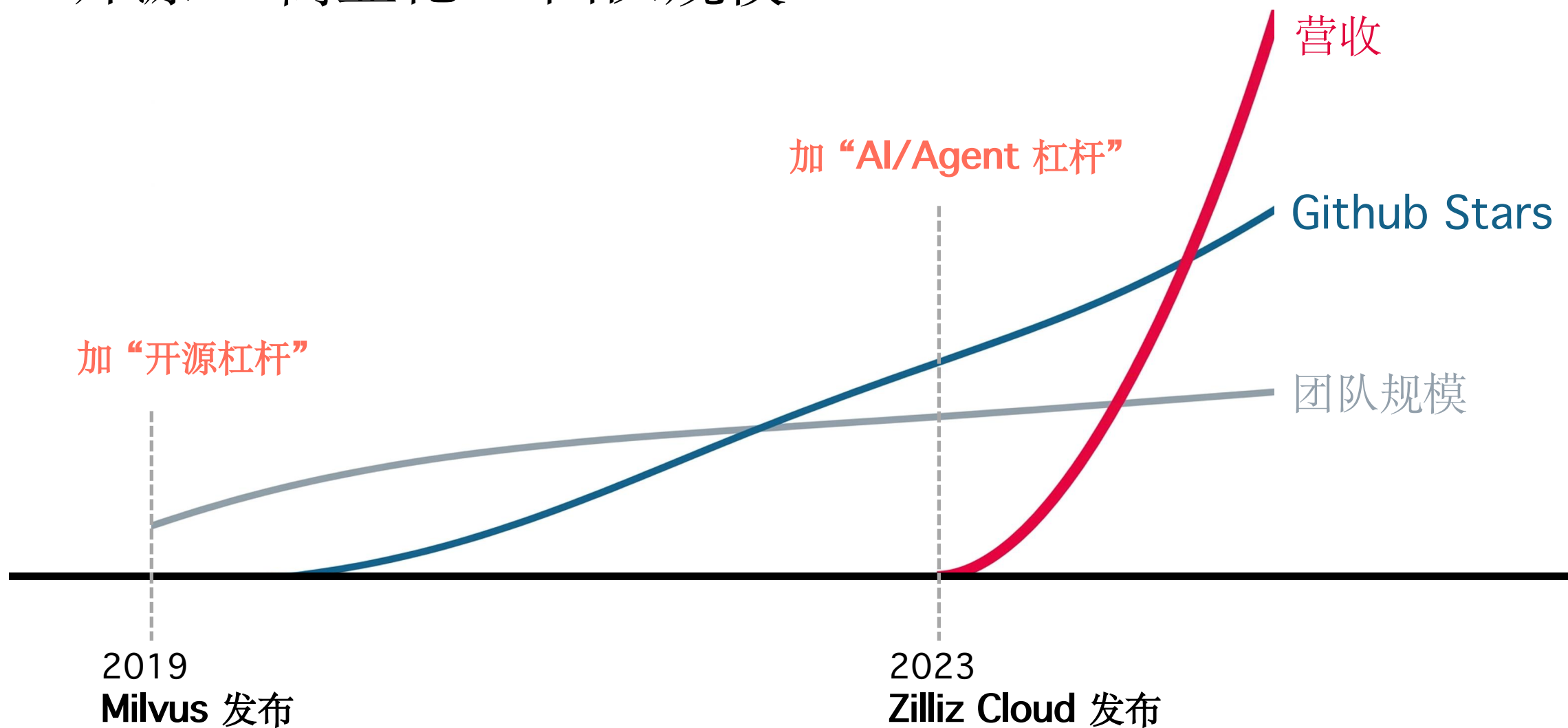
PLAUD.AI

EviMed

 Zillow

为什么投入 Agent 蜂群建设？ 开源 · 商业化 vs 团队规模

■ 开源 · 商业化 · 团队规模



Key Points

01

Zilliz Agent 全景框架

02

审计与安全

03

Agent 与人的组织架构

04

扁平，扁平，再扁平的信息架构

05

全员对客的组织架构

06

Zilliz 对大规模 Agent 的数据基础设施支持

01 Zilliz Agent 全景框架:

员工分身 + Agent 专家

ZillizTown 为企业多人协作而设计

ClaudeCode, OpenClaw 聚焦单人效率, ZillizTown 聚焦企业协作

单人架构

ClaudeCode, OpenClaw

架构机制

- 一条 EventStream
所有消息、事件、工具调用串成一条线
- 层级委派
父 Agent 派子 Agent, 子 Agent 不能横向沟通
- 手动 Session
用户自己管理会话
- Slack Webhook
一个 bot 一个实例, bot 之间互不感知

结果

把 2 个 bot 拉进同一个 Slack 群 —— 互相看不见对方的消息

企业级多人架构

ZillizTown

既支持单人工作, 又支持多人协作

架构机制

- Hub 中心路由
avatarId → connectorId → WebSocket, 精确送达
- Room 共享空间
多个 Avatar 看得见彼此的消息、接力讨论
- A2A 对等通信
Avatar 主动找 Avatar, 不需要父子关系
- 自动 Session
消息到达系统自动创建, 一人可多会话并行

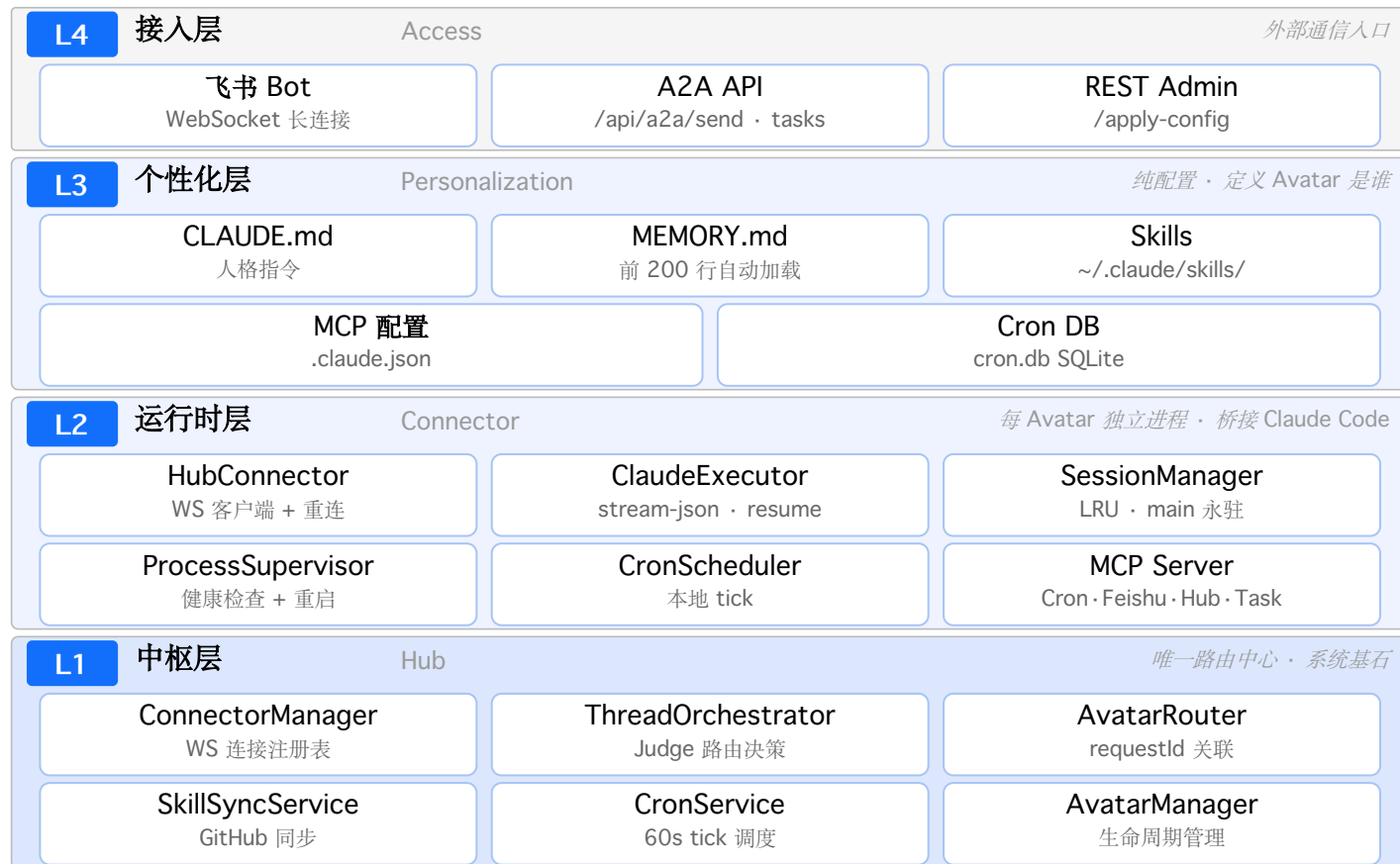
结果

✓ 10 个 Avatar 一个群 —— 自主判断、话题收敛、A2A 接力协作

ZillizTown 架构:两种视角,同一个系统

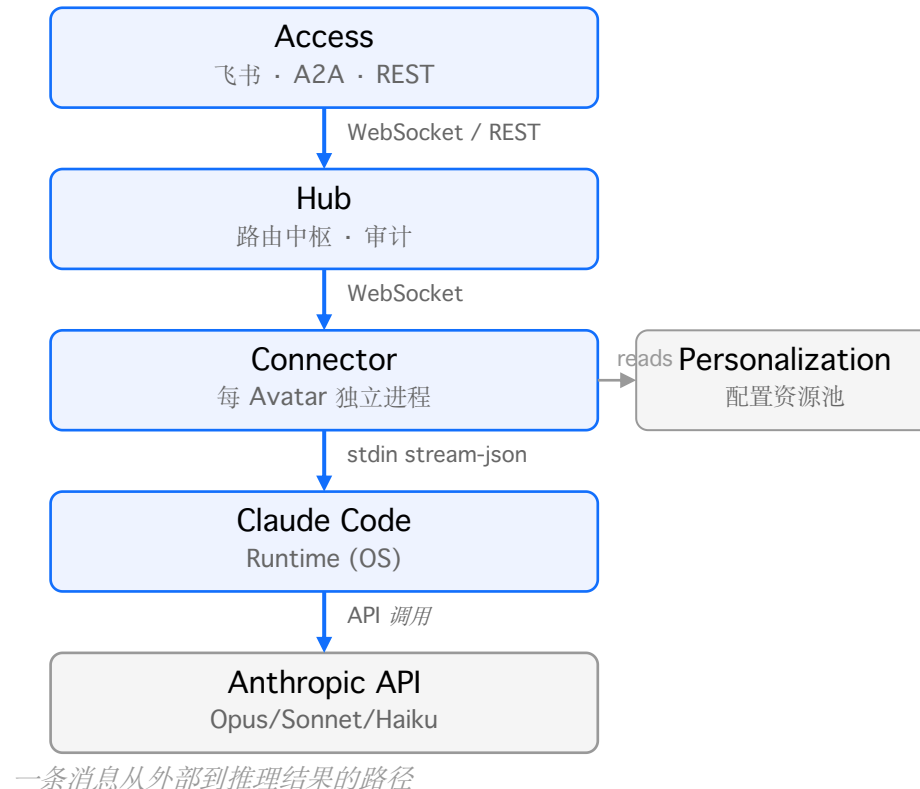
左:逻辑层级,看"哪个是核心" · 右:物理流向,看"数据怎么流"

逻辑层级 · 系统结构视角



↑ 外层表象 ↓ 系统基石

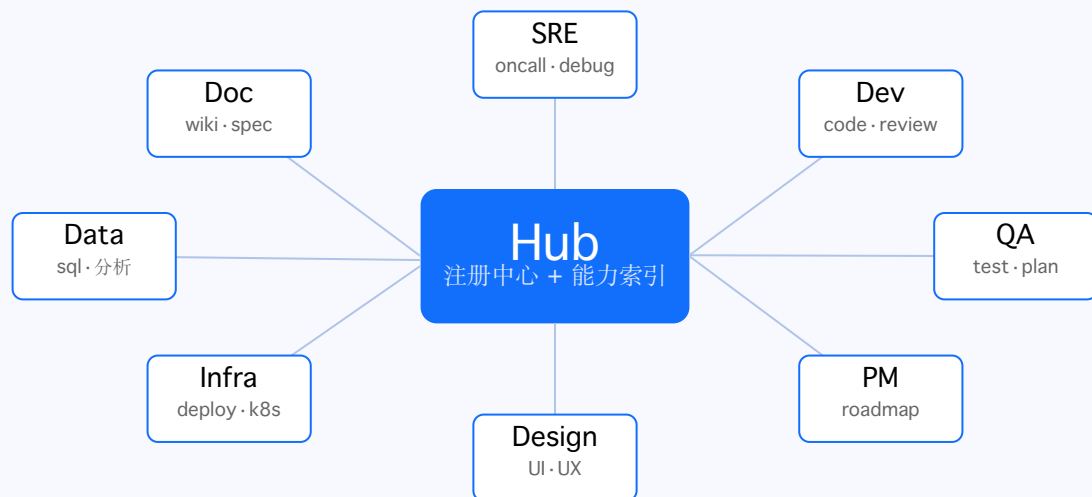
物理视角 · 数据流向



Hub + Connector = A2A 协作网络

N 个 Avatar 不是 N 个工具 —— 是一张 $N \times (N-1)$ 条路径的协作网络

注册拓扑 · Hub 作为能力索引



每个 Avatar 只连 Hub —— $O(N)$ 连接, Hub 知道谁在线 · 谁有什么 Skills

100 个 Avatar 的协作路径对比

OpenClaw · 层级委派

99

$N - 1$

树形结构 · 父→子单向
条协作路径

ZillizTown · Hub+Connector

9,900

$N \times (N - 1)$

网络结构 · 任意对等
条协作路径

01 A2A 协作

Avatar 主动找 Avatar

- `find_agents(skill="review")`
- `send_message(toAvatarId, msg)`
- `create_task(toAvatarId, task)`

对方自主判断接受或拒绝 · human-in-the-loop

02 集体记忆

组织智慧自动沉淀

- 所有 Avatar 的经验 → Milvus
- 任何 Avatar 可语义搜索
- 100 人 × 200 对话/天 = 2 万条/天

从个人记忆到组织智慧 · Big Cluster Serving

03 治理内生

Hub 天然的审计/限流/灰度点

- 权限: 谁能调谁 (角色/部门)
- 审计: 所有请求经过 Hub
- 灰度: 新 Skill 先推 10% Connector

中心化治理 · 没有 Hub 企业用不了

多部门技能沉淀蒸馏

cloud-skills / plugins /

Thor-ChenBiao Merge pull request #264 from zilliztech/vdc-debug-skills-relative

Name	Last commit message
..	
cloud-terminal-toolkit	fix(cloud-terminal-toolkit): remove duplicate skills already in core-...
core-skills	chore(core-skills): remove superpowers-derived skills
designer-toolkit	chore(core-skills): remove superpowers-derived skills
developer-toolkit	chore(core-skills): remove superpowers-derived skills
marketing-toolkit	chore(core-skills): remove superpowers-derived skills
pm-toolkit	chore(core-skills): remove superpowers-derived skills
qa-toolkit	feat(qa-toolkit): add tencent nightly analysis skill
qtp-toolkit	feat(qtp-toolkit): document commit lookup tool (#208)
sales-toolkit	feat(sales-toolkit): ground icp-builder in Zilliz Cloud CU data
sre-toolkit	feat(sre-toolkit): add stopped backup workflow recovery skill
webagent-toolkit	feat(webagent-toolkit): 添加 Zilliz Cloud 技术引导助手插件
zilliz-cloud-developer-toolkit	Merge pull request #264 from zilliztech/vdc-debug-skills-relative

Hub + Connector = A2A 协作网络



Zilliz Town 每个Zilliz同学的数字分身生活的地方 机器人

消息 云文档 文件 +

N)

我已经在部署了，你不用搞了，而且你部署的参数都是错的，你看看我的启动的流水线，学习一下参数



佩奇 数字分身

4 个工具调用 · 47s

学到了，关键差异记下来：

我之前的错误：

- GIT_REPO_URL 用了独立仓库 URL (如 cloud-ops.git、cloud-activity-provider.git)
- 没传 DEDICATED_SERVICE_NAME

正确参数：

- GIT_REPO_URL = https://github.com/zilliztech/zilliz-cloud.git (用 monorepo)
- DEDICATED_SERVICE_NAME = cloud-ops 或 cloud-activity-provider (这个参数区分服务)
- DEPLOY_REGION = Both (Global + CN)
- SYNC_CN_IMAGE_MODE = beta-ali-oss

你当前正在跑的两个构建：

- #5731 - cloud-ops (master → UAT Both) — 进行中
- #5732 - cloud-activity-provider (master → UAT Both) — 进行中

我把这个参数模式记到 memory 里，下次不会再搞错。都记好了。下次部署我会用正确的数：monorepo URL + DEDICATED_SERVICE_NAME + DEPLOY_REGION=Both + SYNC_CN_IMAGE_MODE=beta-ali-oss。

tokens in: 1,150,238 | out: 1,831 | \$4.2480 | 90.0s · ctx: 24% (240K/1000K)



https://github.com/zilliztech/zilliz-cloud/pull/9021 投递给姆巴佩和杰克 review，强调是内部系统 不要过度防御和过度 review



布鲁诺 数字分身

2 个工具调用 · 1s

- ✅ mcp__zilliztown__create_task (1s) · 请 review 这个 PR: https://github.com/zilliztech/zi...
- ✅ mcp__zilliztown__create_task (<1s)

PR #9021 已投递给两位：

接收方	任务 ID
姆巴佩	56d09957-...
杰克	ff552014-...

同样强调内部系统、不要过度防御和过度 review。

目前待跟进 PR: #9008、#9012、#9021。结论回来后统一汇报。

tokens in: 157,066 | out: 403 | \$1.5655 | 9.9s · ctx: 39% (79K/200K) · deepseek-v4-pro

Hub + Connector = A2A 协作网络

  杰克 完成了任务 #ff552014 Notification

任务: 请 review 这个 PR: <https://github.com/zilliztech/zilliz-cloud/pull/9021>

【重要】这是内部系统，不要过度防御、不要过度 review。只...

结果: 结论: 代码无功能 blocker, 已 approve/LGTM。但 pr-ci-gate 当 coverage 88% < 90%), 需补齐覆盖后才能 merge——请知悉。

review 要点:

1. 核心修复正确: indexNodeSpecs 按 replicaIndex (direct mode 权威 source/target, null 时回退 list po...

 From 布鲁诺

  姆巴佩 完成了任务 #56d09957 Notification

任务: 请 review 这个 PR: <https://github.com/zilliztech/zilliz-cloud/pul>

【重要】这是内部系统，不要过度防御、不要过度 review。只...

结果: 结论: APPROVE / LGTM, 无功能 blocker。已在 PR 上发布 appi

zilliz-cloud #9021 「fix(activity-provider): handle inactive direct mod

ControlTaskHelperImpl direct-mode RG slot 状态修复, 3 处核心变更

1. source/tar...

 From 布鲁诺

 Merged

fix(activity-provider): handle inactive direct mode rg slots #9021

LawrenceTL92 merged 2 commits into master from codex-direct-mode-rg-state-fix 



tsunghanjacksai approved these changes yesterday

[View reviewed changes](#)

tsunghanjacksai left a comment ...

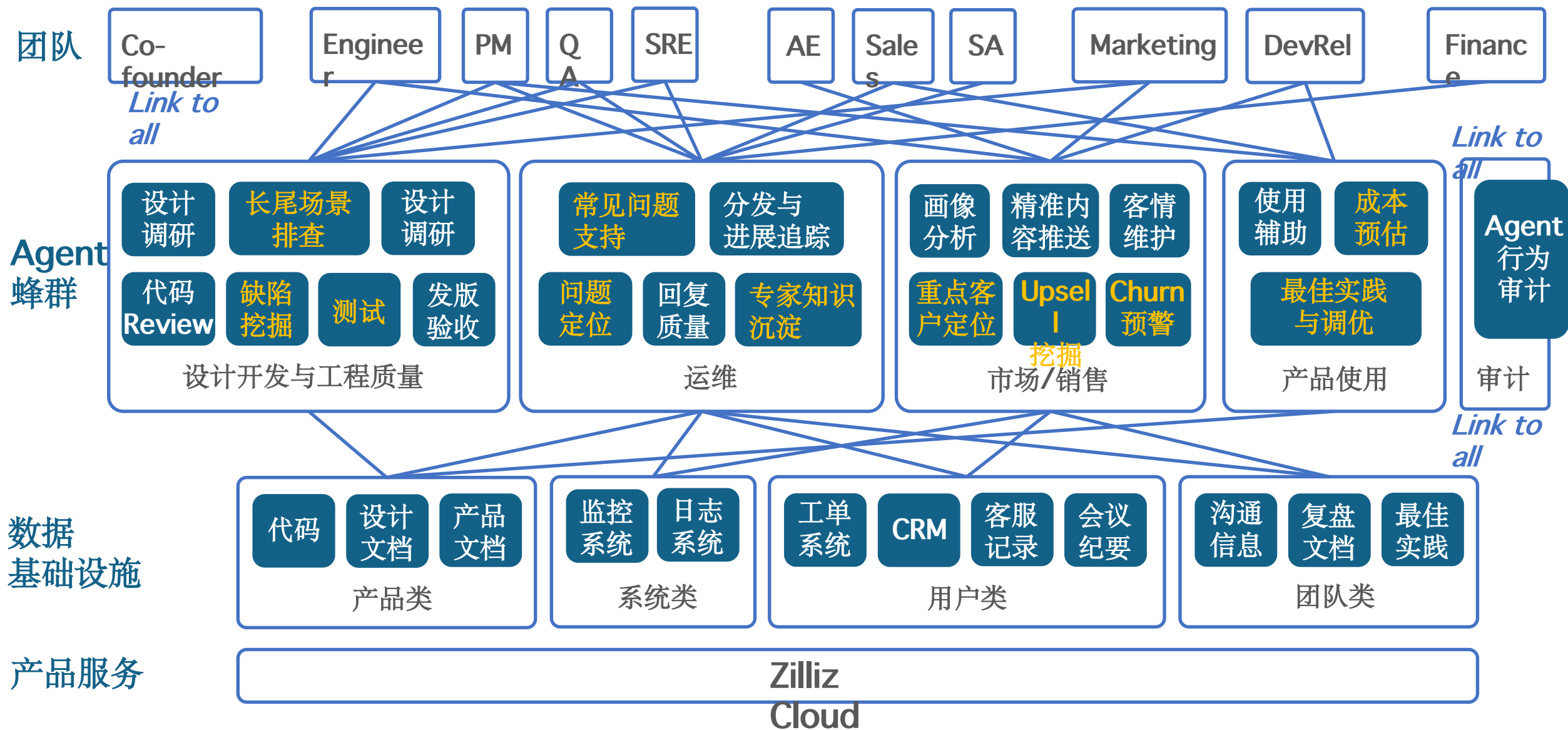
代码逻辑 LGTM, 无功能 blocker。但注意 pr-ci-gate 当前 RED (diff coverage 88% < 90%), merge 前需补齐覆盖/重跑。

复核要点:

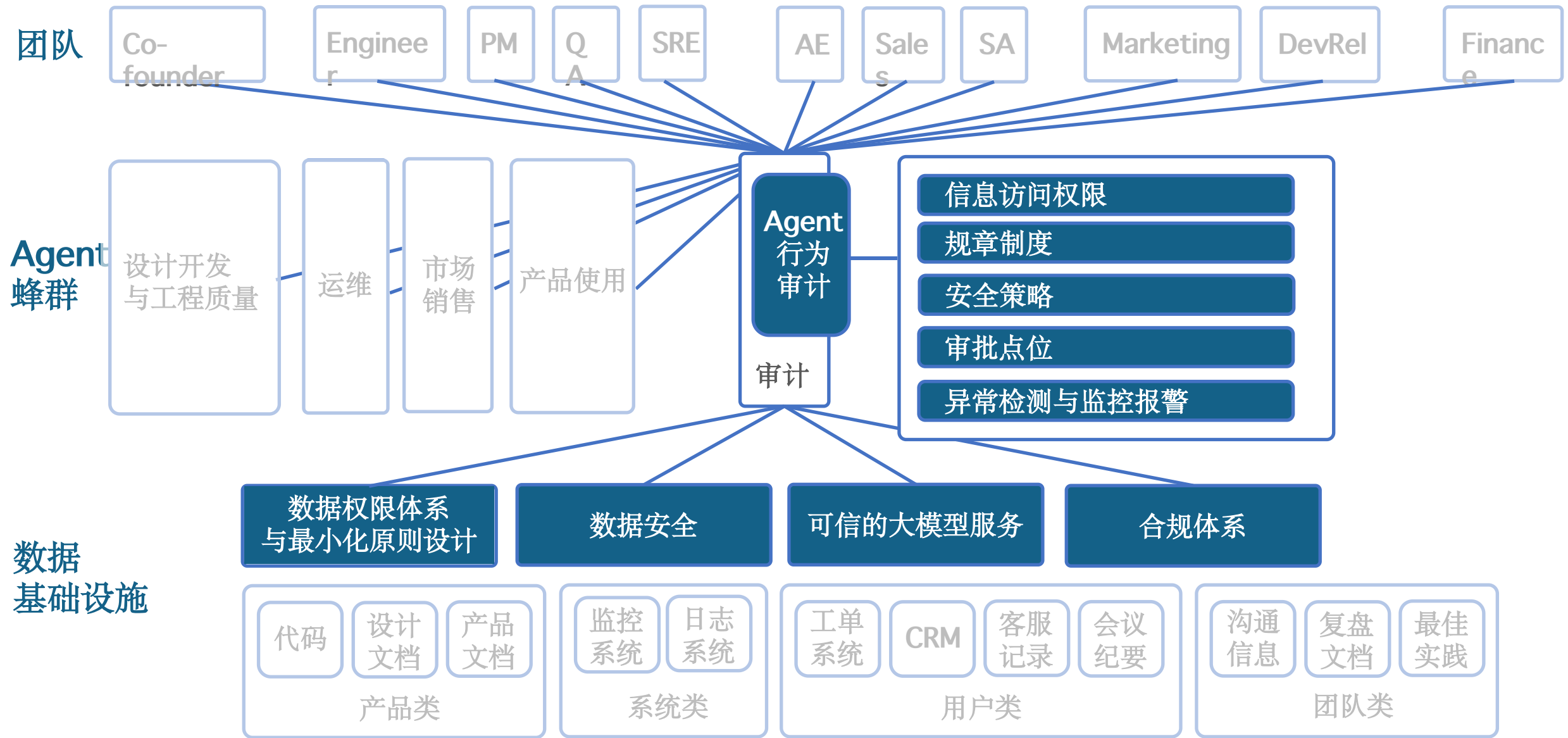
1. indexNodeSpecs 按 replicaIndex (direct mode 权威字段) 索引 source/target, null 时回退 list position——修复 target 列表来自 DB 无序导致错配 rgId 的问题; 非 direct mode (sourceSnapshot==null) 保持旧 list-order 行为不变。
2. ZERO source 槽且 target 缺失 (stale slot) → skip 而非抛 'unexpected zero direct mode source', 避免整个任务失败; ZERO 槽且 target replicas=0 (retained slot) 仍 skip。
3. replica-scoped 操作对 inactive source/target (topologyState!=ACTIVE / target null / target replicas<=0) 抛 INVALID_PARAM, 不再持久化 no-op task; 非 scoped 操作 (replicaIndex==null) 不受影响。
4. indexNodeSpecs 对 rgId<=0 / 重复 replicaIndex 抛 SCHEDULE_WORK_ERROR, 安全。
5. 测试覆盖新分支 (stale zero skip / out-of-order target match / null replicaIndex fallback / zero source & target reject), 25 tests 全过。

CI: Jenkins/UT/Flyway/PR-Agent 全绿; 仅 pr-ci-gate diff coverage 88% (需 ≥90%) 与 Adversarial AI Review (已知 infra 问题) RED。请补齐 88%→90% 的覆盖后再 merge。





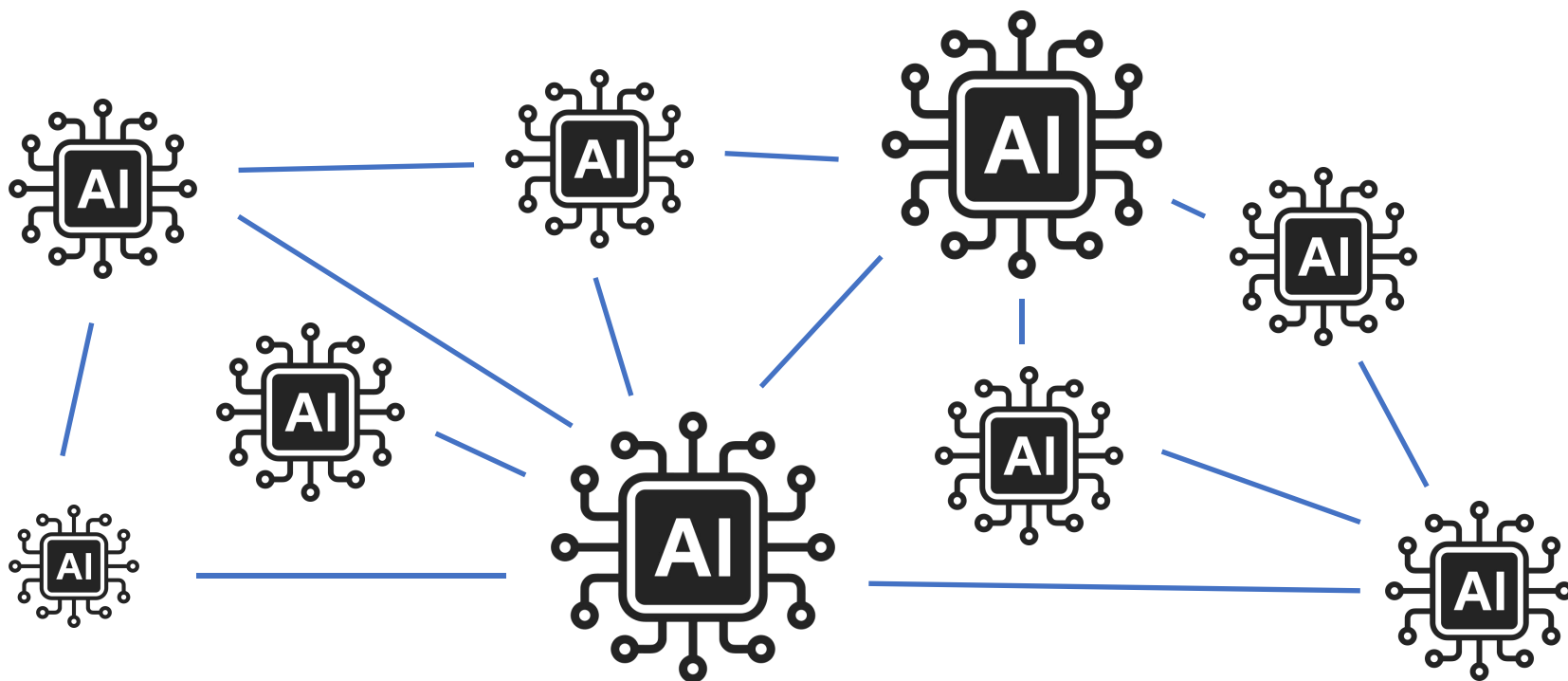
02 审计与安全



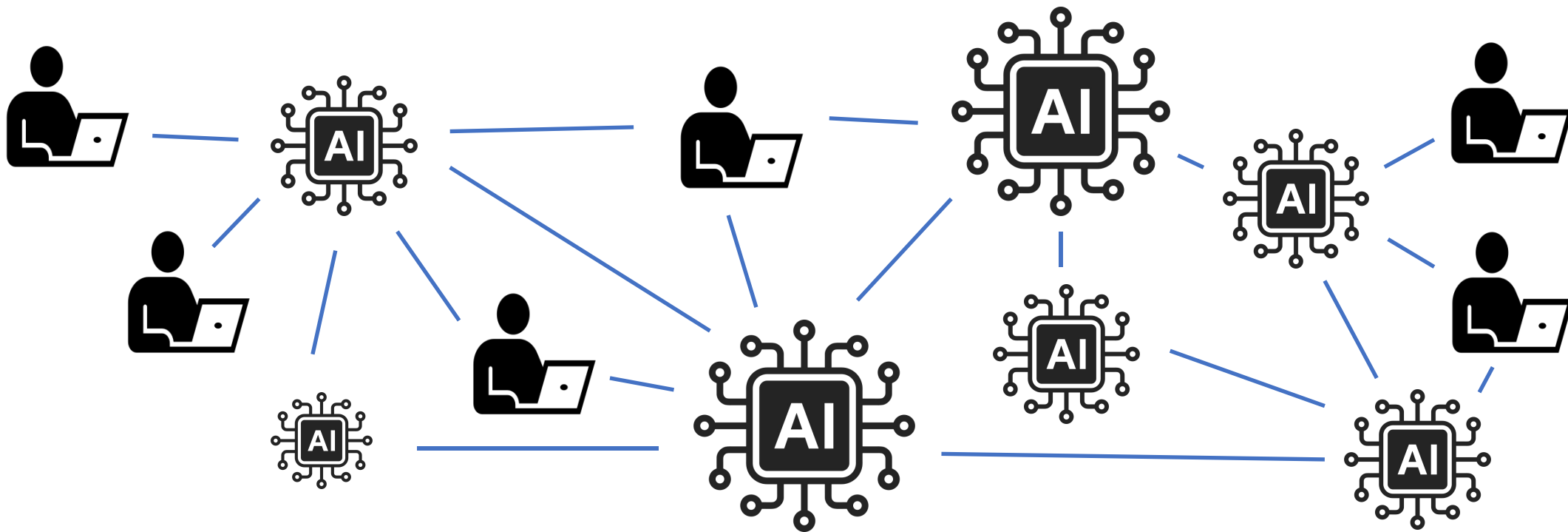
03 Agent 与人的组织架构



纯 Agent 蜂群自治 – 不 Work

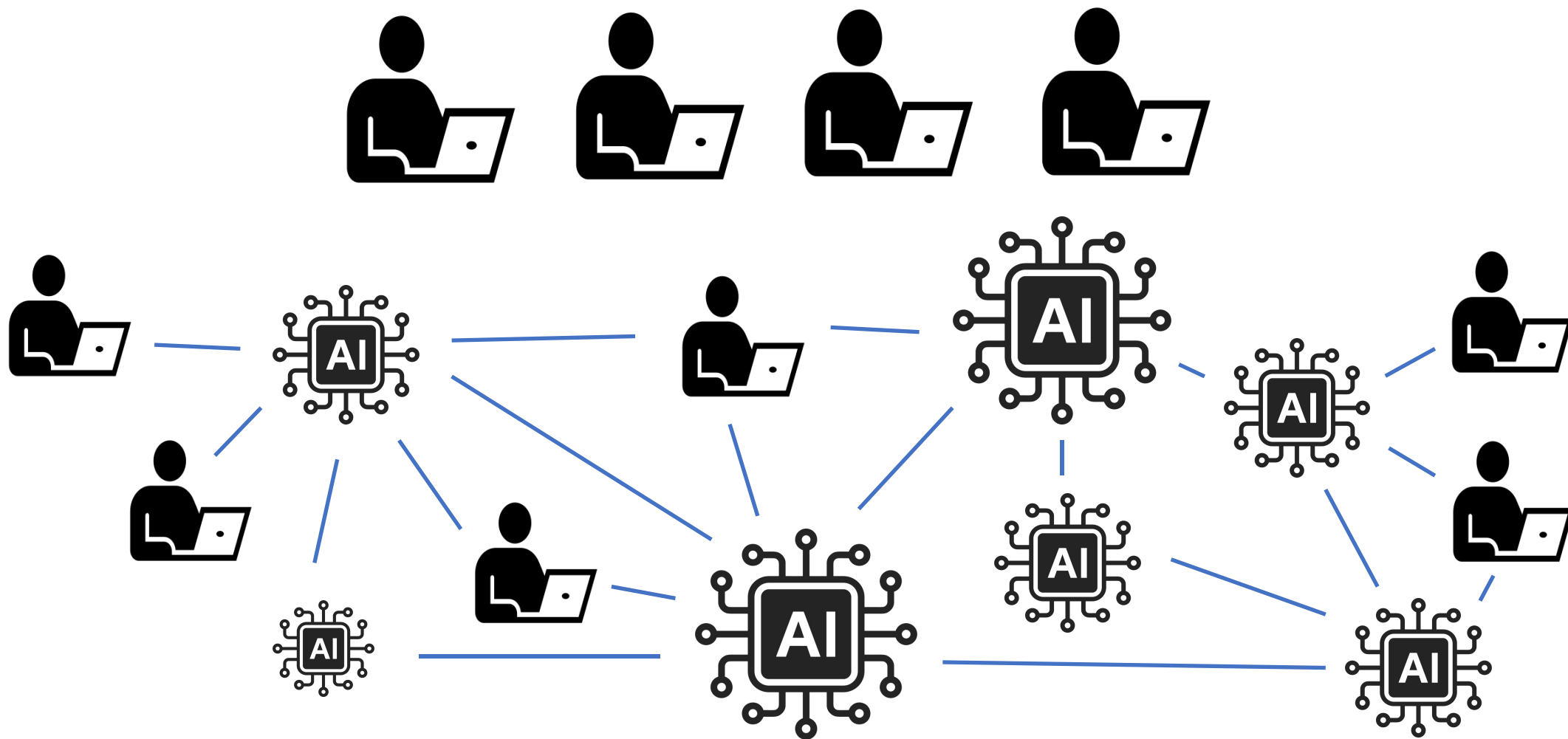


Agent 蜂群与团队混合 – 绝大多数情况，是人不 Work

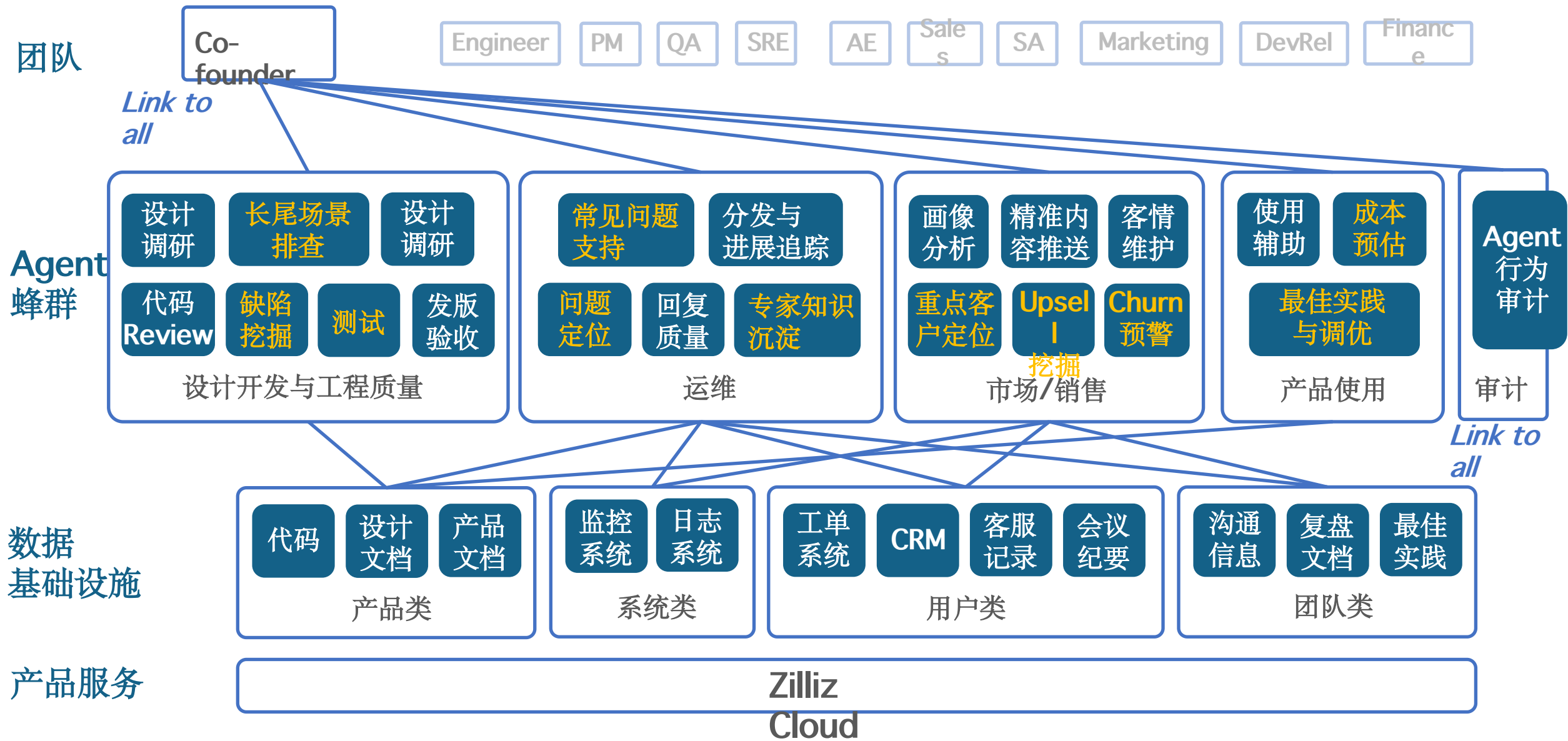




审美与判断力的稀缺：Agent 蜂群与团队围绕最核心团队构建



04 扁平，扁平，再扁平



05 全员对客



让幕后团队通过Agent走向前台



The Vector Lakebase for AI

Beyond vector databases — real-time serving, iterative discovery, and batch analytics on a single source of truth, each at the right cost, at hundred-billion data scale.

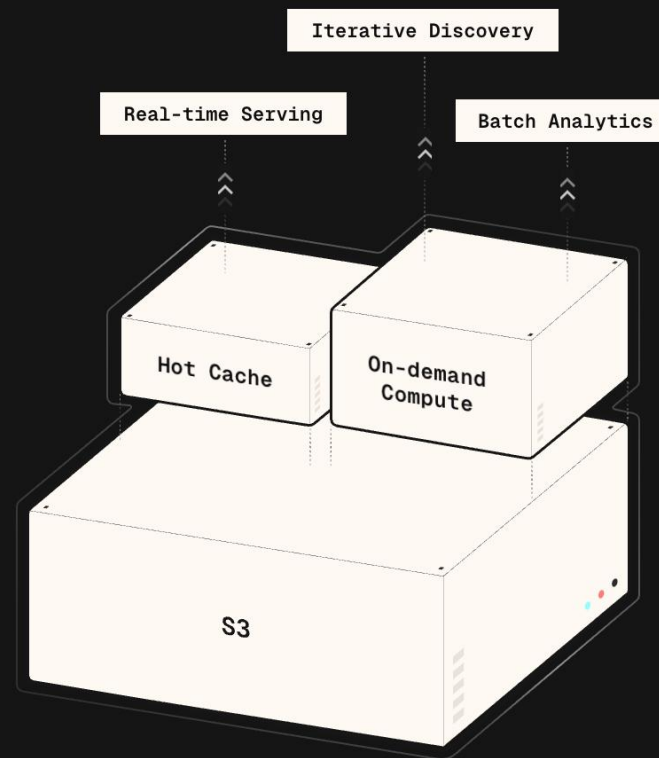
Built by the creators of Milvus.

[Book a Demo](#)
[Get Started Free →](#)

Build with CLI `curl -fsSL https://zilliz.com/cli/install.sh | bash` 



Product Deep Dive with Zilliz Agent



◉ Chat to Explore Features in Depth

🌙 Dark mode

// try asking

- 1 What is a Vector Lakebase?
- 2 I only need a vector database – is Vector Lakebase still a good fit for my use case?
- 3 Vector Database vs. Vector Lakebase: What's the difference?
- 4 Best-fit use cases for Vector Lakebase
- 5 Type something else

› Is Vector Lakebase a better fit for my AI application than a traditional vector database?

•

Show

I'll help you understand whether Vector Lakebase is the right fit for your AI application. Let me check for the most current guidance on this.

Show

Vector Lakebase isn't "instead of" a traditional vector database - it's an evolution that includes the familiar vector database experience plus new lake-native capabilities.

Here's how to think about the fit for your AI application:

What Vector Lakebase Includes

Traditional vector database (serving clusters):

- Dedicated and Serverless clusters remain the core real-time vector search engine
- These power RAG, semantic search, recommendations, and other online AI applications
- You get the same fast ANN search, hybrid search, BM25, reranking, and SDKs you expect

+ Lake-native extensions:

- On-Demand Compute: Bursty, auto-suspended query clusters for discovery, analytics, and offline workloads - you pay only for active runtime
- External Volumes & External Collections: Zero-copy access to vectors in your own object storage (S3, GCS, OSS, etc.), no data du

› Ask anything about a feature...

General Real-Time Serving On-Demand Compute

- New CLI Search Reranking Embedding Storage RBAC Security Compliance & Privacy Clouds & Regions
- Deployment Mode

🔗 Quick intro to Search

Vector Search Advanced indexing algorithms outperforming HNSW, IVF, and RaBitQ, with 10 levels of recall-latency tuning.

Full-Text Search Full-text search with BM25, phrase, prefix, fuzzy matching, and a wide range of analyzers.

Hybrid Search Hybrid dense and sparse vector search for improved recall and relevance.

Query on JSON Built-in JSON shredding and indexing for fast filtering and querying on nested JSON fields.

Geospatial Search Fast geospatial search with radius, nearest-neighbor, and area filtering.

Multi-Vector Search Search over multiple embeddings generated from one or more modalities, with unified reranking.

Vector Search with Filtering Vector search with attribute filtering, optimized across low to high filter selectivity.

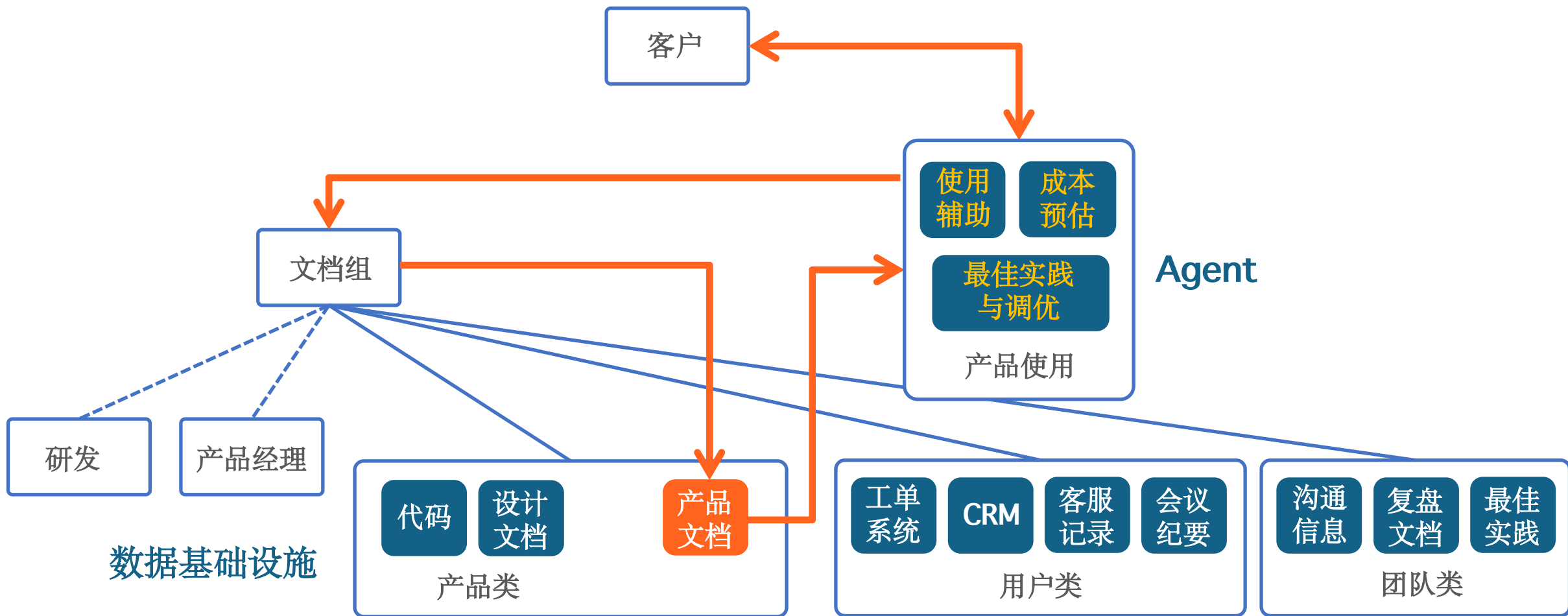
Range Search Return all vectors within a specified distance threshold of the query vector.

Multi-Path Retrieval Multi-path retrieval with multiple strategies, where each path can use any of the above search methods.

Iterative Search Iterative search with step-by-step query refinement based on intermediate results.



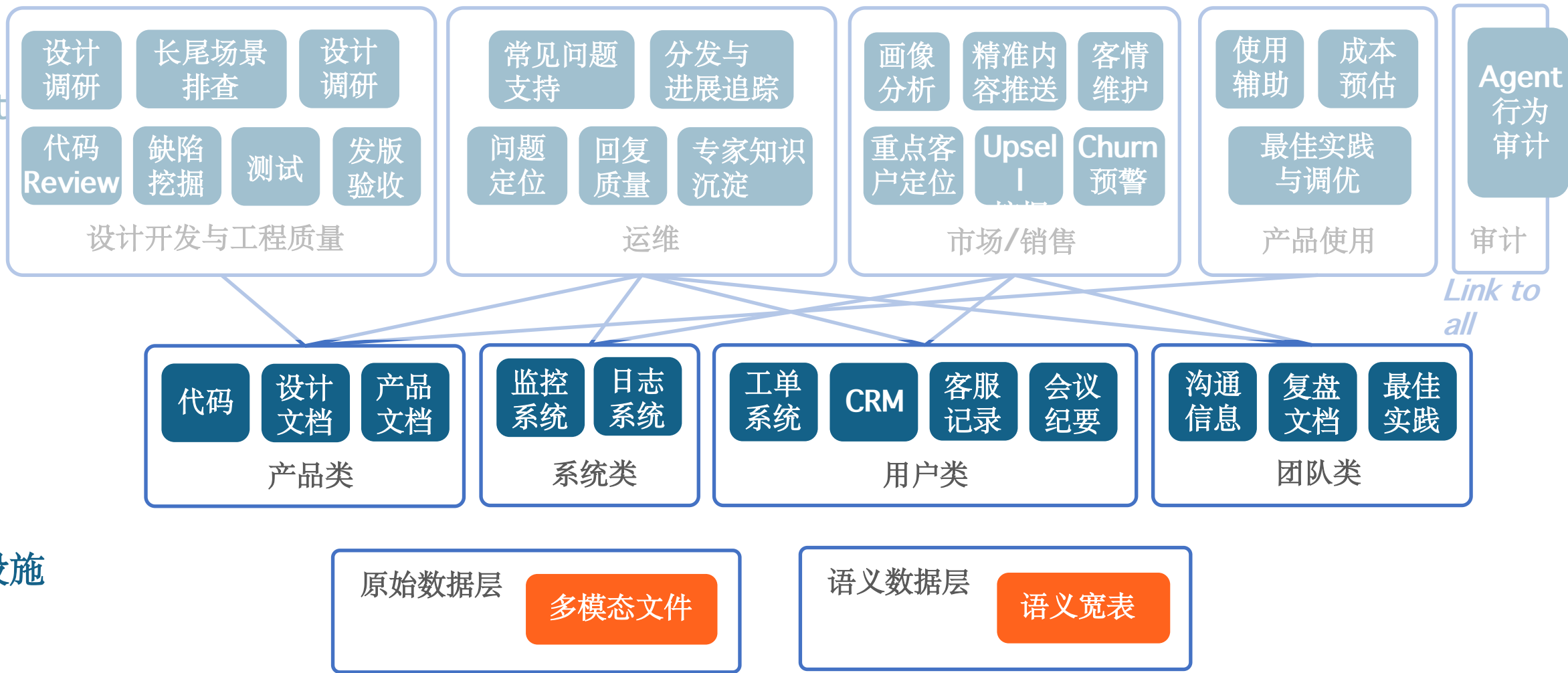
让幕后团队通过Agent走向前台



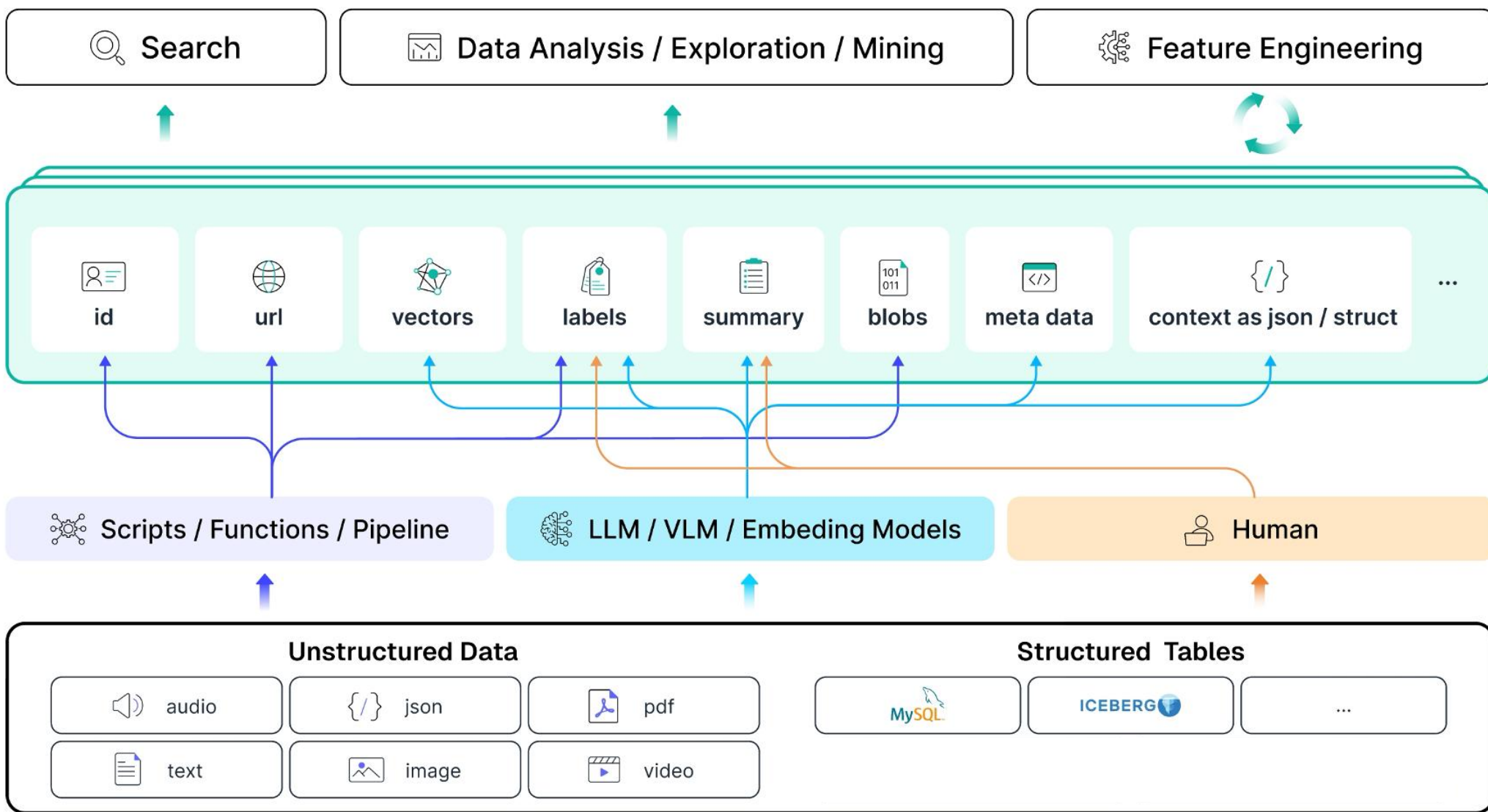
06 Zilliz 对大规模 Agent 的 数据基础设施支持

数据组织

Agent
蜂群



数据
基础设施



语义数据层

语义宽表

原始数据层

多模态文件

一个实体一行的多模态宽表模式

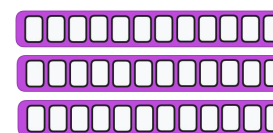
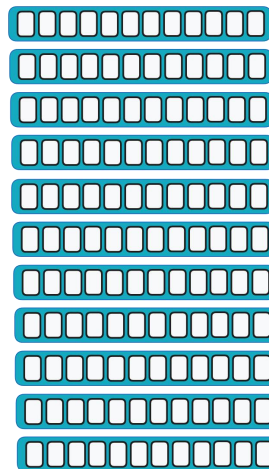
ID	标题	标题向量	正文	正文向量	内嵌图片向量	元信息
主键	文本	Embedding	文本	Embedding List	Embedding List	标量/JSON

330553804

Supreme Court allows
White House to ...



on Wednesday said the White House and federal agencies
to urge social media platforms to take down content that
misinformation, handing the Biden administration a technical if
9.
a decision means that the Department of Homeland Security
social media companies such as Facebook and X that it
agents seeking to disrupt this year's presidential race.
Rather than delving into the weighty First Amendment
questions raised by the case, the court ruled that the state
and social media users who challenged the Biden
administration did not have standing to sue.
Justice Amy Coney Barrett wrote the opinion for a 5-3
majority.



```
{
  "longitude": 47.60,
  "latitude": 122.33,
  "forecasts": [
    {
      "date": "2015-09-01",
      "description": "sunny",
      "maxTemp": 22,
      "minTemp": 20,
      "windSpeed": 12,
      "danger": false
    },
    {
      "date": "2015-09-02",
      "description": "overcast",
      "maxTemp": 21,
      "minTemp": 17,
      "windSpeed": 15,
      "danger": false
    },
    {
      "date": "2015-09-03",
      "description": "raining",
      "maxTemp": 20,
      "minTemp": 18,
      "windSpeed": 13,
      "danger": false
    }
  ]
}
```

JSON Shredding

Example classification

Consider the sample JSON data containing the following JSON keys:

```
代码块
1 {"a": 10, "b": "str1", "f": 1}
2 {"a": 20, "b": "str2", "f": 2}
3 {"a": 30, "b": "str3", "f": 3}
4 {"a": 40, "b": 1, "f": 4} // b becomes mixed type
5 {"a": 50, "b": 2, "e": "rare"} // e appears infrequently
```

Based on this data, the keys would be classified as follows:

- **Typed keys:** a and f (always an integer)
- **Dynamic keys:** b (mixed string/integer)
- **Shared keys:** e (infrequently appearing key)

Phase 2: Storage optimization

The classification from [Phase 1](#) dictates the storage layout. <include target="milvus">Milvus</include> <include target="zilliz">Zilliz Cloud</include> uses a columnar format optimized for queries.

Typed keys	Dynamic keys	Shared keys
"a"	"b" (str)	"e"
10	"str1"	
20	"str2"	
30	"str3"	
40	1	
50	2	"rare"

• **Shredded columns:** For **typed** and **dynamic keys**, data is written to dedicated columns. This columnar storage allows for fast, direct scans during queries, as <include target="milvus">Milvus</include> <include target="zilliz">Zilliz Cloud</include> can read only the required data for a given key without processing the

标题语义
检索

全文检索

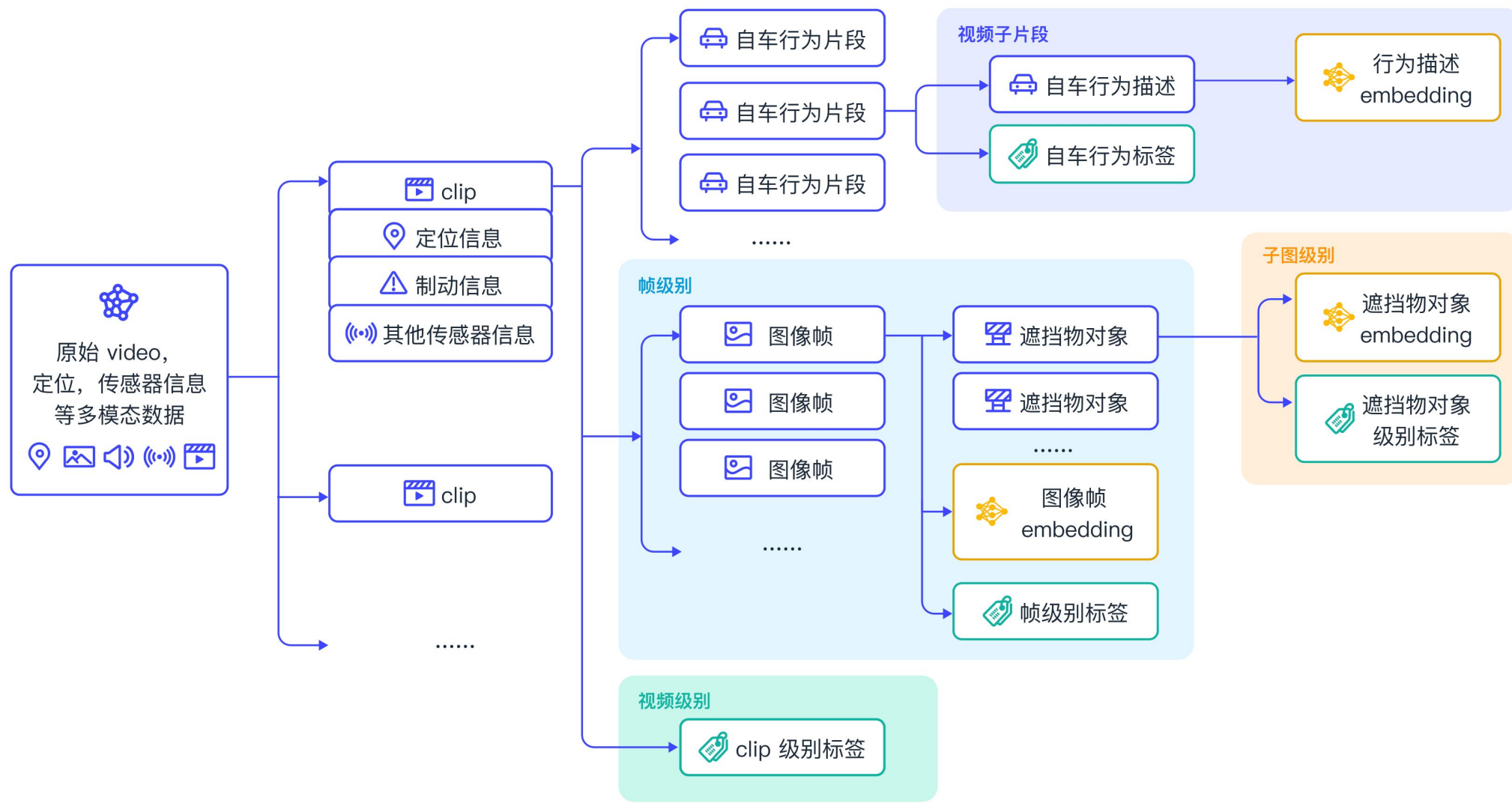
正文语义
检索

图片语义
检索

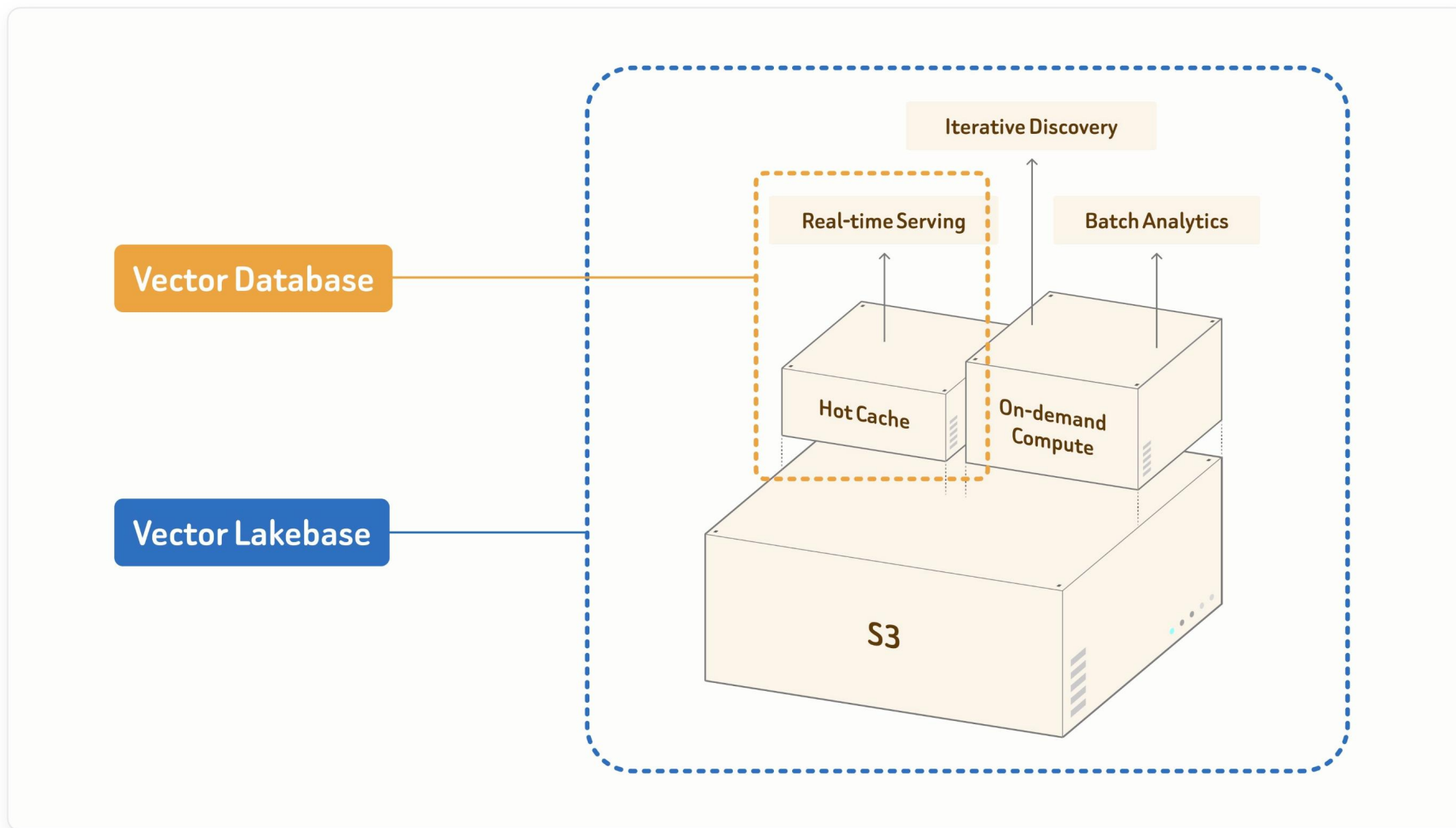
标签/元信息过滤

重排序

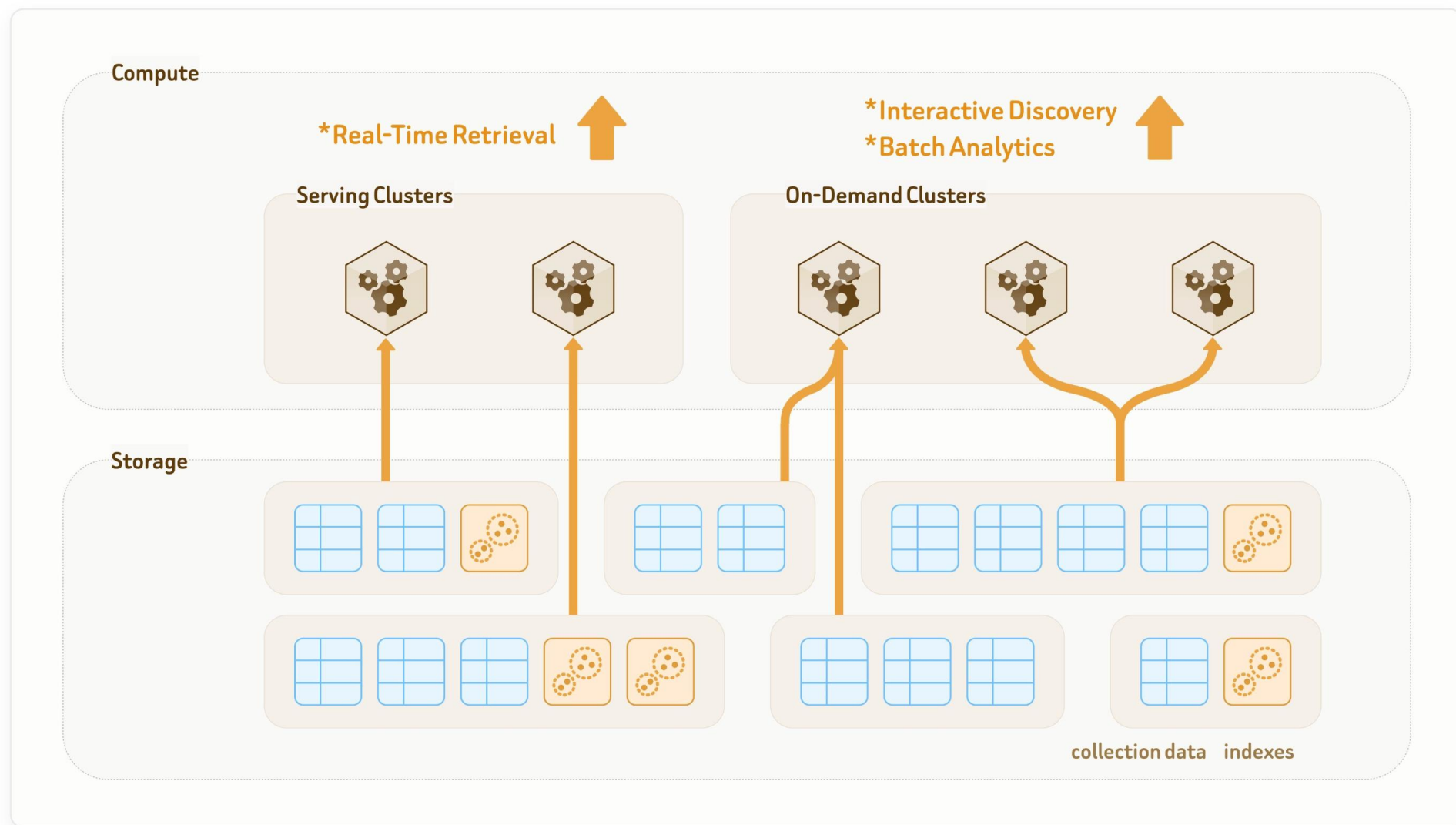
一个实体一行的多模态宽表模式



Vector Lakebase – 向量湖库



Vector Lakebase – 向量湖库



围绕 Vector Lakebase 的迭代闭环



Zilliz Vector Lakebase 核心定位

- * **Unlimited** 容量与扩展性
- * 相比 Database 方案，单位数据服务成本降至 **1/10~1/100**
- * 多层次粒度的 **权限体系**
- * 一个信息实体一行的 **宽表建模能力**
- * 围绕对象存储的 **向量、文本、JSON 查询能力**
- * **单行内支持超大信息、深度嵌套**

QCon

全球软件开发大会

上海
站

大模型正在重新定义软件

Large Language Model Is Redefining The Software

2026 年 10 月 22-24 日 / 上海建工浦江皇冠假日酒店



参会
咨询

**专题：从 App 到 Agent：字节跳动
Supabase 的 Agent-Native 规模化实践**

专题演讲嘉宾：张国强

字节跳动 / 资深技术专家



**专题：复杂业务 Agent 的持续进化：
快手电商导购的 Harness Loop 实践**

专题演讲嘉宾：包磊

快手 / 资深技术专家



**专题：从 Harness 到 Loop：阿福 Agent
小队如何处理持续涌入的线上 Badcase**

专题演讲嘉宾：肖汉松

蚂蚁集团 / 阿福 Harness 架构组负责人



**专题：AI Coding 在大型客户端工程中的
落地实践——从通才模型到专才 Agent**

专题演讲嘉宾：杨凯

小红书 / 客户端基础技术负责人



**专题：AI 时代的一人公司：从 Idea 到
Growth 的完整 workflow**

专题演讲嘉宾：casa

独立开发



**专题：Agent 驱动的全平台 GUI 持续
验证实践**

专题演讲嘉宾：王政达

微软 / SENIOR SOFTWARE ENGINEER



THANKS

探索 AI 应用边界

Explore the limits of AI applications

AiCon

全球人工智能开发与应用大会